

DESCRIPTION

ALZ 3720 is a general purpose submersible hydrostatic level transmitter. High-quality silicon piezoresistive sensor, robust stainless steel housing and a hydrometric cable with the ventilation tube make ALZ 3720 a reliable and cost-effective solution for a wide variety of industrial level measurement applications.

An integrated Pt100 temperature sensor is optionally available.

SPECIFICATIONS

Level: 1 mH₂O up to 250 mH₂O

Basic accuracy: $\pm 0.25\%$

Probe: Ø27 mm, AISI 316L stainless steel

Outputs: 4...20 mA; 0...20 mA; 0...10 V; 0...5 V; RS-485 / Modbus RTU; HART®

Sensor: silicon piezoresistive

Media temperature: -20...+70 °C

Optional: Ex ia, Pt100 temperature sensor

HART® is a registered trademark of the HART Communication Foundation.

APPLICATIONS

Water, wastewater

Fuels, oils

Tanks, wells

Pump protection

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar	Level, mH ₂ O	Overpressure, bar	Pressure range, bar	Level, mH ₂ O	Overpressure, bar
0...0.1	0...1	1	0...2.5	0...25	6
0...0.16	0...1.6	1	0...4	0...40	15
0...0.25	0...2.5	1	0...6	0...60	15
0...0.4	0...4	1	0...10	0...100	30
0...0.6	0...6	3	0...16	0...160	60
0...1	0...10	3	0...25	0...250	60
0...1.6	0...16	6	-	-	-

PERFORMANCE	P > 0.4 bar	P ≤ 0.4 bar
Accuracy, % of span*	≤ ±0.25 (standard) / 0.2 (optional)	≤ ±0.5 (standard)
Temperature effect (% of span / 10 °C)	≤ ±0.1	≤ ±0.2
Compensated range	-20...+70 °C	0...+70 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms with analog output, ≤ 200 ms with digital output	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-20...+70 °C		
Cable service temperature (static applications)	PVC:	-20...+70 °C	
	PUR:	-50...+70 °C	
Storage temperature	-20...+70 °C		
Approval	0Ex ia IIC T6...T4 Ga X		
Temperature class	T4	T5	T6
Ambient temperature	-20...+70 °C	-20...+60 °C	-20...+50 °C
Vibration resistance	10 g RMS, 25–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10 ⁶ cycles		

MECHANICAL SPECIFICATIONS

Housing	stainless steel 316L (1.4404)
Seal	FKM; EPDM; NBR
Diaphragm	stainless steel 316L (1.4435)
Cable external jacket	PVC, PUR, FEP
Ingress protection	IP69K

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U _s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	≤ [(U _s – 12 V) / 0.02 A] Ohm*	≤ 26 mA
4...20 mA / HART®			
4...20 mA / 3-wire		≤ 500 Ohm	
0...20 mA / 3-wire		≥ 10 kOhm	< 7 mA
0...10 V / 3-wire			
0...5 V / 3-wire	5 V	≥ 5 kOhm	≤ 2 mA
0.5...4.5 V / 3-wire	6...15 V		≤ 7 mA
0.5...4.5 V / 3-wire	12...36 V		
RS-485 / Modbus RTU		-	

* For output signal 4...20 mA / HART® minimum load resistance for digital communication: 250 Ohm.

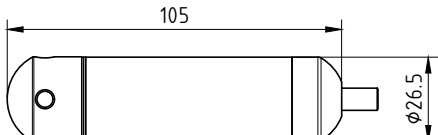
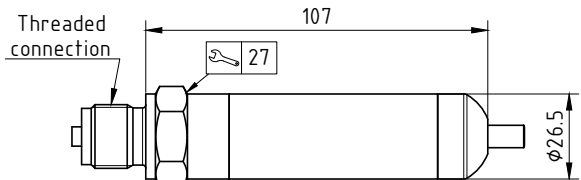
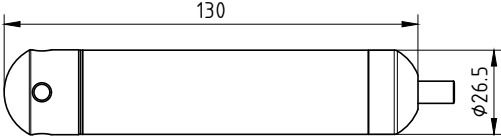
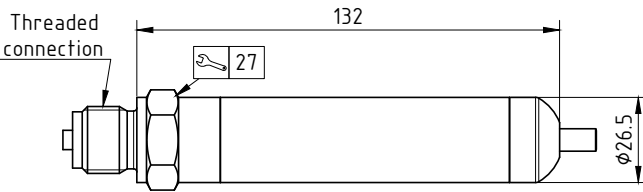
Safe values for intrinsically safe design 0Ex ia IIC T6...T4 Ga X:

Parameter	2-wire	3-wire
Maximum voltage, U _i	28 V	6 V
Maximum current, I _i	93 mA	60 mA
Maximum power, P _i	660 mW	100 mW
Maximum internal inductance, L _i	10 µH	10 µH
Maximum internal capacitance, C _i	15 nF	500 nF

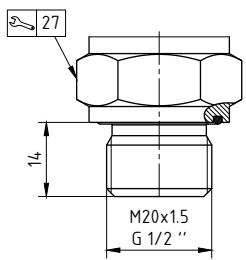
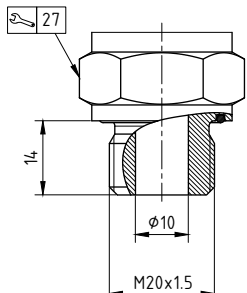
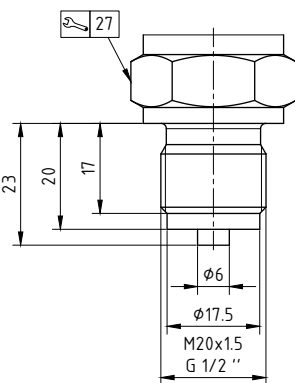
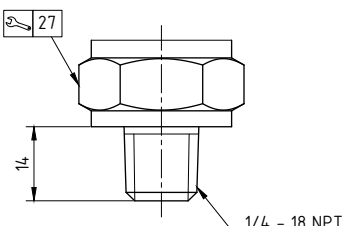
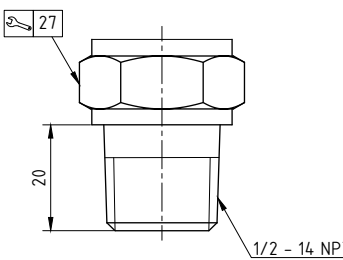
ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits		Cable gland wires
2-wire	power +	white (red)
	power -	brown (blue)
	ground	yellow-green
3-wire	power +	white (red)
	power -	brown (blue)
	signal +	green (black)
	ground	yellow-green
Pt100 (optional)	T+	yellow
	T-	pink (white)
	T-	gray (green)
RS-485	power +	white (red)
	power -	brown (blue)
	A	yellow
	B	green (black)
	ground	yellow-green
HART®	power +	white
	power -	brown
	ground	yellow-green

DIMENSIONS (mm)

Standard version	Threaded version G1/2 EN; Code 73
	
with HART or Ex ia	with HART or Ex ia
	

THREADED CONNECTIONS (NIPPLES), DIMENSIONS (mm)

M20x1.5 DIN 3852; Code 20 G1/2" DIN 3852; Code 72	M20x1.5 DIN 3852 open port; Code 26	M20x1.5 EN 837; Code 21 G1/2" EN 837; Code 73
		
1/4" NPT; Code 74	1/2" NPT; Code 82	
		

ORDERING CODE

		-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
UNIT OF MEASUREMENT									
	bar	B							
	kg/cm ²	S							
	mH ₂ O	W							
	kPa	K							
	Other (specify when ordering)	X							
UPPER RANGE LIMIT (URL)									
bar, kg/cm ²		mH ₂ O		kPa					
0.1	0100	1	1000	10	1001				
0.16	0160	1.6	1600	16	1601				
0.25	0250	2.5	2500	25	2501				
0.4	0400	4	4000	40	4001				
0.6	0600	6	6000	60	6001				
1	1000	10	1001	100	1002				
1.6	1600	16	1601	160	1602				
2.5	2500	25	2501	250	2502				
4	4000	40	4001	400	4002				
6	6000	60	6001	600	6002				
10	1001	100	1002	1000	1003				
16	1601	160	1602	Other	XXXX				
25	2501	250	2502						
Other	XXXX	Other	XXXX						
ACCURACY									
						0.25% (P > 0.4 bar) (standard)	C		
						0.5% (P ≤ 0.4 bar) (standard)	D		
						0.2% (P > 0.4 bar)	B		
						Other (specify when ordering)	X		
OUTPUT SIGNAL									
						4...20 mA / 2-wire (standard)	A		
						4...20 mA / 2-wire, 0Ex ia IIC T6...T4 Ga X	Q		
						4...20 mA / 3-wire	B		
						0...20 mA / 3-wire	C		
						0...10 V / 3-wire	D		
						0...5 V / 3-wire	E		
						0.5...4.5 V / 3-wire, U _s = 5 V, 0Ex ia IIC T6...T4 Ga X	R		
						0.5...4.5 V / 3-wire, U _s = 6...15 V	K		
						RS-485 / Modbus RTU	M		
						4...20 mA / HART®	H		
						Other (specify when ordering)	X		
SEALS									
						FKM	F		
						NBR	N		
						EPDM	E		
						Other (specify when ordering)	X		
VERSION									
						Standard	00		
						Pt100 temperature sensor	0T		
						Threaded version, M20x1.5 DIN 3852	20		
						Threaded version, M20x1.5 EN 837	21		
						Threaded version, M20x1.5, open port	26		
						Threaded version, G1/2" DIN 3852	72		
						Threaded version, G1/2" EN 837	73		
						Threaded version, 1/4" NPT	74		
						Threaded version, 1/2" NPT	82		
						Threaded version G1/2 EN	NC		
						Other (specify when ordering)	XX		

ORDERING CODE (CONTINUED)

ALZ 3720	-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
CABLE EXTERNAL JACKET								
							PVC	P
							PUR	U
							FEP	T
							Other (specify when ordering)	X
CABLE LENGTH								
Any length in meters (e.g., 010M means 10 meters)								XXXM

ACCESSORIES

				
Anchoring clamp for submersible transmitter	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		